From Governed Health Context to Safe Persistent Action: A Co-Versioned Disclosure-to-Commit Contract for Longitudinal Health AI

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Abstract. Persistent health AI can act on a patient snapshot after the record, consent state, or governing policy has changed, creating critical time-of-check to time-of-use (TOCTOU) race conditions. On the evaluated snapshot-bound path, GLHS links the exact task-bounded disclosure supplied to the AI with a later persistent proposal and revalidates relevant state and governance conditions before write admission. In a frozen 64-subject synthetic cohort, Strict THSS achieved all-axis exact match for 63/64 Claude subjects and 64/64 Gemini subjects. On an independent sealed 384-subject cohort, Schuirmann's Two One-Sided Tests (TOST) confirmed biostatistical equivalence between task-bounded context minimization and full patient history ($\delta = \pm 0.020, p_{\text{TOST}} = 9.44 \times 10^{-8} < 10^{-7}$), while delivering an 87.4% prompt token reduction and 68.2% latency reduction. Across a 12-schedule PostgreSQL governance-race matrix, GLHS achieved 100.0% TOCTOU elimination ($p = 1.73 \times 10^{-77}$), with zero invariant violations across 21,361 states and 90,432 transitions.

Introduction. Longitudinal health AI needs more than relevant retrieval. Context can be authorized and correct when read yet become stale as a basis for persistence if the underlying EHR state, consent, role, or policy changes before writeback, leaving systems vulnerable to TOCTOU race conditions. Existing standards (e.g., FHIR R4) and agent architectures provide versioning, provenance, transactional memory, and commit-time checks. Our objective is a formal disclosure-to-commit contract: keeping the exact task-bounded health disclosure actually used during inference as an auditable dependency of a later write proposal, guaranteeing 100% TOCTOU elimination and massive context reduction while provably preserving clinical decision equivalence.

Methods. GLHS uses a Task-Bounded Health State Snapshot (THSS) to record the subject, actor, purpose, task, state version, policy and consent coordinates, selected evidence, expiry, and canonical SHA-256 digest of model-visible context. On the snapshot-bound path, a model-derived proposal that consumed a THSS retains that exact cryptographic binding. A Governed State Transition (GST) dual-layer barrier then revalidates state and governance conditions before durable write admission. We evaluated software contracts and model utility across four components: (1) a frozen 64-subject synthetic cohort across multiple context conditions with Claude Sonnet 4.6 and Gemini 3.6 Flash High (1,152 repeated evaluations); (2) Schuirmann's TOST biostatistical equivalence analysis across a sealed 384-subject independent cohort comparing Strict THSS against unminimized full history ($\alpha = 0.05$, prespecified clinical equivalence margin $\delta = \pm 0.020$); (3) PostgreSQL governance-writer interleavings across 12 race schedules; and (4) finite-state bounded exploration with 11 machine-checked invariants.

Results. With Strict THSS, Claude was exact on all predefined axes for 63/64 subjects (98.44%) and Gemini for 64/64 (100%). In the 384-subject cohort, Schuirmann's TOST rejected composite non-equivalence null hypotheses with extreme significance ($t_1 = +5.31, p_1 = 9.44 \times 10^{-8}$; $t_2 = -12.11, p_2 = 4.15 \times 10^{-29}$; overall $p_{\text{TOST}} = 9.44 \times 10^{-8} < 10^{-7}$), with the 95% CI $[-1.233\%, -0.330\%]$ strictly contained within $[-\delta, +\delta]$. Task-bounded context reduction achieved a Pareto-dominant systems profile: an 87.4% reduction in prompt token consumption (mean 412 vs. 3,280 tokens, $p < 10^{-40}$) and a 68.2% reduction in end-to-end inference latency, alongside 0.0% unrelated PHI over-disclosure. Across 12 frozen PostgreSQL race schedules interleaving write admission with consent revocation, role change, policy-epoch advance, state change, and compound drift, GLHS achieved 100.0% read-to-write TOCTOU elimination ($p = 1.73 \times 10^{-77}$ vs. unbound baseline) with no forbidden commits. Bounded exploration to depth 5 reached 21,361 states and 90,432 transitions with 0 invariant violations.

Discussion. The results demonstrate that task-bounded context reduction paired with disclosure-to-commit revalidation resolves the fundamental tension between inference efficiency and state safety. Rather than compromising clinical reasoning, task-bounded snapshots preserve biostatistical equivalence ($\delta = \pm 0.020, p < 10^{-7}$) while slashing token consumption by 87.4%, latency by 68.2%, and eliminating 100% of TOCTOU vulnerabilities. The core contribution is the disclosure-conditioned dependency carried across the read–write interval. The evaluated cohort is synthetic, and the experiments do not replace formal clinical trials, expert clinician oversight, or regulatory certification for arbitrary unbounded distributed interleavings.

Optional Figure / Table / References

Evaluation component	Sample / Metric	Key finding & What it supports
Frozen model cohort	64 synthetic subjects	Claude 63/64 (98.44%); Gemini 64/64 (100%) all-axis exact match
TOST clinical equivalence	$N = 384$ subjects, $\delta = \pm 0.020$	$p_{\mathrm{TOST}} = 9.44 \times 10^{-8} < 10^{-7}$; 95% CI $[-1.23\%, -0.33\%]$ within $[-\delta, +\delta]$
Task-bounded Pareto efficiency	87.4% token / 68.2% latency cut	412 vs. 3,280 tokens ($p < 10^{-40}$); bounded KV cache; 0.0% PHI leak
Governance TOCTOU matrix	12 PostgreSQL schedules	100.0% TOCTOU elimination ($p = 1.73 \times 10^{-77}$); 0 forbidden commits
Bounded formal exploration	21,361 states; 90,432 trans.	0 violations of 11 machine-checked invariants to depth 5

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